Team A - Ames Housing Project Plan

Project Overview

In this project, we're taking a deep dive into housing market data to predict residential property values using the Ames Housing Dataset. Following Doris's feedback, our main goal is to build and compare linear models against ensemble methods to see which performs best.

- **Dataset Scope:** We're working with the Ames Housing Dataset, which includes 1,460 home sales records and around 79 features covering all kinds of property details.
- **Core Objective:** Predict home `SalePrice` values as accurately as possible through careful model comparison.
- **ML Strategy (Doris's Feedback):** We're setting up a baseline with Linear Regression and testing it against a Random Forest Regressor to see if an ensemble model captures non-linear relationships better and boosts overall accuracy.

Methodology

Our workflow covers the full data lifecycle, taking raw SQL database records through Python analysis and modeling all the way to an interactive dashboard.

- **Data Extraction & Ingestion:** Use SQL queries to pull raw data, filter relevant subsets, join tables, and check the database schema.
- **Exploratory Data Analysis (EDA):** Explore features in Python using pandas, NumPy, seaborn, and matplotlib to spot potential outliers, check for multicollinearity, and decide on transformations.
- **Feature Engineering & Preprocessing:** Clean up missing values, apply one-hot and ordinal encoding, and fix skewed variables for linear and tree models.
- **Model Development & Evaluation:** Train, tune, and cross-validate Linear Regression and Random Forest models with scikit-learn, tracking performance using RMSE and R^2.
- **Business Intelligence & Reporting:** Build a clean Power BI dashboard to share key findings, walk stakeholders through scenarios, and showcase model metrics.

Task Breakdown

Welcome to our practical task guide! We've built up great momentum over the last six weeks. Let's collaborate, support each other when debugging gets tough, and split up these tasks based on what skills everyone wants to focus on.

SQL Tasks (Data Extraction):

- **Task 1:** Write a SQL query to review the table structure and pull all residential property records, filtering out non-standard sales (like setting `SaleCondition = 'Normal'`).
- **Task 2:** Create a query using GROUP BY to summarize total home counts, average sale price, and median price per square foot across each neighborhood.
- **Task 3:** Write a query to locate NULL or missing values across key columns (like LotFrontage, GarageType, and basement metrics) so we know what to clean in Python.

Python Tasks (EDA - Exploratory Data Analysis):

- **Task 1:** Check the distribution, skewness, and kurtosis of `SalePrice`, plotting a histogram with a KDE curve to see if log transformation makes sense.
- **Task 2:** Identify the top 5 neighborhoods with the highest median home values and create a horizontal bar chart with seaborn to visualize them.
- **Task 3:** Plot living area size (GrLivArea) against `SalePrice` on a scatter plot to catch clear outliers (such as huge properties selling unusually cheap).
- **Task 4:** Build new features (like TotalSqFt = TotalBsmtSF + 1stFlrSF + 2ndFlrSF, and HouseAge = YrSold - YearBuilt) and handle missing categorical entries by filling NAs with 'None'.

Machine Learning Tasks (Prediction):

- **Task 1:** Build a baseline model for `SalePrice` using Linear Regression on an 80/20 train-test split, relying on RMSE as our primary metric (along with R^2) to heavily penalize larger pricing mistakes.
- **Task 2:** Train a Random Forest Regressor, tuning parameters like n_estimators=100 and max_depth=10, to see how well tree ensembles handle non-linear trends and cut down test RMSE compared to the baseline.
- **Task 3:** Summarize evaluation metrics (MAE, RMSE, R^2) in a side-by-side table; keeping RMSE front and center;and pull out the top 10 most influential features using feature_importances_.

Power BI Tasks (Dashboarding):

- **Task 1:** Load the cleaned dataset and model outputs into Power BI, then create basic DAX measures for Average Sale Price, Total Houses Sold, and Prediction Error (Actual - Predicted).
- **Task 2:** Add a Clustered Bar Chart on Page 1 displaying Average Sale Price by Neighborhood, along with an Overall Quality score slicer (ranging from 1 to 10).
- **Task 3:** Create a Scatter Chart on Page 2 plotting Actual vs. Predicted Sale Price, including a 45-degree reference line to quickly evaluate model performance.

Project Execution Plan & Roadmap

Building upon the foundational work established during our Task breakdown, our team executed an expanded analytics and modeling workflow to advance the Ames Housing predictive modeling project.

Completed Milestones

- **Advanced SQL Data Preparation:** Refined extraction scripts to clean missing features, isolate single-family housing records, and construct structured views for automated ingestion into Python environment pipelines.
- **EDA & Statistical Validation:** Executed rigorous statistical testing, evaluated skewness across target pricing distributions, applied log transformations, and pinpointed key influential variables.
- **Model Benchmarking:** Trained and tuned baseline Linear Regression alongside Random Forest Regressor models using cross-validation, achieving measurable RMSE improvements and establishing key feature importances.
- **Dashboarding & Stakeholder Delivery:** Designed and deployed interactive Power BI dashboards displaying key DAX metrics, error tracking, and valuation breakdowns by neighborhood and property quality attributes.

Achieved Outcomes

- **Predictive Accuracy Gain:** Reduced evaluation error metrics across test sets, demonstrating clear performance gains over standard linear baseline models.
- **End-to-End Pipeline Integration:** Successfully established a reproducible workflow connecting database queries directly to model evaluation and BI visual reporting.

SQL Analysis Summary

Here is a summary of our team's SQL work on the Ames Housing Dataset, detailing what we accomplished, who contributed what, how the experience felt, how we worked together, and our key takeaways.

What We Did

We started by setting up an SQLite database and loading in data straight from Pandas DataFrames. After checking the schema to understand the fields and data types, we filtered for Residential Low-Density homes sold under Normal conditions. We then wrote aggregation queries to summarize home counts and average prices by neighborhood, and ran targeted checks to find missing values in critical columns.

Contributions

Everyone pitched in across the board;from setting up the initial database environment and checking schemas to crafting filter queries and writing aggregate functions for neighborhood summaries.

The Experience

Working through SQL hands-on gave us great practical experience. Moving from raw database setup to writing and tweaking queries showed us just how crucial solid data validation is before jumping into modeling.

Teamwork

Collaboration made a huge difference. We bounced query ideas off one another, tackled complex joins and grouping logic together, and worked through debugging bugs as a team whenever errors popped up.

Lessons Learnt

- **Technical Skills:** Got comfortable setting up SQLite databases, pushing Pandas DataFrames into SQL tables, writing aggregate queries, and checking data for missing values systematically.
- **Teamwork Skills:** Improved our technical communication, worked better together, practiced patience during bug hunts, and made key project decisions as a group.

Conclusion

Our SQL exploration gave us a solid data foundation and clean dataset, paving the way for smooth feature engineering and predictive modeling in Python.

Exploratory Data Analysis and ML Summary

This section recaps our main takeaways, statistical techniques, and machine learning concepts from the Python EDA and modeling work.

Exploratory Data Analysis (EDA)

- **Research Questions & Feature Engineering:** We framed clear research questions and engineered relevant features (like calculating delivery delays from timestamps) to uncover meaningful patterns in our data.
- **Summary Statistics & Visualization:** We used Pandas and Seaborn to compute summary stats (mean, median, IQR) and created charts (histograms, boxplots) to inspect skewed distributions and spot extreme outliers.

Statistical Testing

- **Hypothesis Testing Selection:** We picked statistical tests based on how our data was distributed, turning to non-parametric options like the Wilcoxon Signed-Rank Test when dealing with heavily skewed variables instead of relying strictly on paired t-tests.
- **Execution & Interpretation:** We ran tests using `scipy.stats`, evaluated p-values against standard alpha levels, and noted potential limitations like omitted variables.

Machine Learning (Theory & Practice)

- **Evaluation Metrics Alignment:** We matched evaluation metrics to the specific problem at hand such as: focusing on Precision for fraud detection to minimize false alarms, or tracking RMSE for continuous predictions like home prices and delivery ETAs to penalize big errors heavily.
- **Model Training & Validation:** We built and evaluated Logistic Regression and Decision Tree models in Scikit-Learn using train/test splits, cross-validation, hyperparameter tuning, and macro F1 scores.

Conclusion

The Ames Housing project brought together everything we learned about using modern analytics tools to solve real valuation problems. By combining SQL, Python, machine learning, and Power BI, we built a complete end-to-end data pipeline that improved our pricing predictions. We identified key structural and geographic factors driving home prices, turning raw data into clear market insights. Overall, this project showed us how a well-structured data workflow leads to better predictive performance and smarter real estate decisions.